

NK Cell Exhaustion in Wilson's Disease Revealed by Single-cell RNA Sequencing Predicts the Prognosis of Cholecystitis

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Abstract

Metabolic abnormalities associated with liver disease have a significant impact on the risk and prognosis of cholecystitis. However, the underlying mechanism is complicated and remains to be elucidated. In particular, the effect of metabolic abnormalities on the progression of cholecystitis through the regulation of immune cell function is poorly understood. In this study, we investigated this issue using Wilson's disease (WD) as a model. Wilson's disease is a genetic disorder characterized by impaired mitochondrial function and abnormal copper metabolism. Our retrospective clinical study of over 600 patients with WD found that they have a significantly higher incidence of cholecystitis and a poorer prognosis. The immune cell landscape in the hepatic mesenchymal stromal microenvironment using single-cell RNA sequencing showed that the tissue immune microenvironment is altered in patients with WD, mainly a major change in the constitution and function of the innate immune system, including enhanced antigen presentation process, activation of the immune response, and activation of lymphocytes. Exhaustion of natural killer (NK) cells is the fundamental factor, supported by an increase in the expression of the inhibitory receptors NKG2A and TIGIT and a decrease in the expression of cytotoxic molecules. Clinical tissue and blood samples verified increased NKG2A⁺ and TIGIT⁺ NK cells and decreased IFN γ ⁺ NK cells in WD. Further bioinformatic analysis has confirmed a positive correlation between NK cell exhaustion and poor prognosis in cholecystitis and other inflammatory diseases. The study demonstrated abnormal function of liver mesenchymal immune cells triggered by specific metabolic dysfunction in WD, with a focus on the correlation between NK cell exhaustion and poor healing of cholecystitis. Our findings highlight the immune cell dysfunction due to metabolic changes in hepatocytes and provide new insights into the improvement of inflammatory diseases by assessing immune cell function.

eLife assessment

The findings are **useful** for understanding the disease's pathology and immune dysregulation, but the evidence is still **incomplete** regarding whether these immune changes are directly caused by copper metabolism alterations or are secondary to liver dysfunction.

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Introduction

Cholecystitis refers to a group of inflammatory diseases with different etiologies and clinical courses¹. While the corresponding pathological mechanisms of cholecystitis are complicated, systemic inflammation is widely recognized as a primary catalyst of cholecystitis pathology^{2–6}. Cholecystectomy is typically the preferred treatment option for patients with severe cholecystitis and gallstones, whereas medication is prescribed for those with mildly symptomatic cholecystitis or calculous cholecystitis. However, the effectiveness of alternative or conservative treatments can vary greatly. Furthermore, a substantial percentage of patients face the possibility of complications or recurrence^{7,8}. Therefore, it is important to assess the prognosis of pharmacological treatment in patients with cholecystitis through prospective evaluation. Inflammatory diseases, ranging from common inflammatory patterns to rare chronic illnesses, are characterized by a defective immune system and an exaggerated inflammatory cascade response. Compelling evidence supports that the proportion, gene expression profile, and functional status of immune cells hold significant prognostic value^{9–14}. Therefore, it is critical to comprehensively comprehend the immune cell dysfunction linked with cholecystitis, its corresponding altered characteristics, and their prognostic implications.

Natural killer (NK) cells are the major innate lymphocyte subset and are found in most organs^{15,16}. They produce lytic granules that eliminate infected or cancerous cells, coordinate the inflammation process, form immunological memory, and modulate antigen-presenting cell function in inflammatory diseases^{17–19}. In the context of tumors and chronic infections, NK cells exhibit an exhausted phenotype, termed NK cell exhaustion, which is characterized by downregulated expression of activating receptors (e.g., NKG2D, CD16), upregulated expression of inhibitory receptors (e.g., NKG2A, PD-1, TIGIT, Tim-3), decreased production of effector cytokines, and impaired cytolytic activity^{20,21}. Recently, blockade of NKG2A and TIGIT has been shown to reverse NK cell exhaustion and enhance the therapeutic effects of anti-tumor and anti-infective immunotherapies^{21–23}. In addition, NK cell exhaustion has been reported to predict poor prognosis in hepatocellular carcinoma and acute myeloid leukemia; however, the correlation of NK cell exhaustion with prognosis in cholecystitis and other inflammatory diseases remains largely unexplored^{24–26}. The liver plays a critical role in immune defense. It contains a substantial population of diverse immune cells, particularly NK cells^{27–29}. The liver modulates these cells to mount immune responses upon encountering specific inflammatory triggers^{28,30}. Furthermore, liver pathologies affect mesenchymal immune cell function by means of regulating intercellular signaling, thereby influencing systemic inflammatory responses. Liver disease commonly impacts the development and outcome of inflammation-associated diseases such as cholecystitis^{31–33}. Wilson's disease (WD) is a rare genetic disorder caused by mutations in the gene encoding the transmembrane copper-transporting P-type ATPase protein³⁴. It leads to deficits in mitochondrial structure and function, as well as specific alterations in liver metabolism, resulting in a variety of liver diseases. Additionally, WD is

commonly associated with inflammatory conditions in the clinic^{35–38}. Thus, WD, with its unique etiology, serves as an excellent model for exploring the regulation of immune cell function by hepatic metabolic abnormalities.

In this retrospective clinical study of over six hundred WD patients, it was found that they had a significantly higher cholecystitis incidence than the general population. This, in turn, is associated with a poor prognosis for cholecystitis in these patients. A thorough examination of the immune cell landscape in the hepatic mesenchymal stromal microenvironment using single-cell RNA sequencing (scRNA-seq) revealed that the tissue immune microenvironment is altered in patients with WD. Specifically, there were significant changes in the composition and function of the innate immune system, including an increase of mononuclear phagocytes (MPs) and NK cells, as well as enhanced pro-inflammatory characteristics. Additionally, there was an enhancement in the biological processes of antigen presentation, activation of immune response, and activation of lymphocytes. One key event among these changes was the exhaustion of NK cells, as demonstrated by an increase in the expression of inhibitory receptors NKG2A and TIGIT and a decrease in the expression of cytotoxic molecules. Multiplex immunohistochemistry and flow cytometry techniques confirmed the exhaustion of NK cells in both clinical liver tissue and peripheral blood samples. Finally, our bioinformatics analysis has confirmed a positive correlation between NK cell exhaustion and poor prognosis in cholecystitis and other inflammatory diseases. Our study has demonstrated abnormal liver mesenchymal immune cell function triggered by specific metabolic dysfunction in WD, focusing on NK cell exhaustion and its correlation with poor healing of cholecystitis. These findings provide foundation for further studies of effect of hepatocytes metabolic changes on immune cell function and insights into improvement of inflammatory diseases by assessing immune cell function.

Results

WD causes poor prognosis of cholecystitis

WD is a genetic disorder resulting from mutations in the *ATP7B* gene, exhibiting significant genetic heterogeneity with over 600 mutations or polymorphisms³⁹. Through sequencing DNA samples from 20 patients with WD, we found that c. 2333G>T (p. R778L) accounted for the majority (65%) of the three most common mutations in Chinese, while c. 2804C>T (p. T935M) and c. 2975C>T (p. P992L) were not detected (**Figure 1a**)⁴⁰. In hepatocytes, *ATP7B* deficiency in the trans-Golgi network (TGN) leads to abnormalities in mitochondrial function and copper metabolism, and ultimately to a wide range of liver diseases (**Figure 1b**)^{41–44}. Typical clinical manifestations are liver fibrosis and Kayser-Fleischer Ring in the cornea that caused by Cu accumulation in the cornea (**Figure 1c**, d)⁴¹. Additionally, metabolic symptoms of WD patients, such as a higher plasma total cholesterol (T-CHOL) concentration and a lower risk of hyperglycemia (Cohort 1 and Cohort 2), are observed (**Figure 1e**, Figure S1a). Mitochondrial stress and glycolysis assays revealed that knockout of the *ATP7B* (*ATP7B*-KO) in human hepatocyte cell lines WRL 68 significantly impaired the basal oxygen consumption rate (OCR) and capacity of ATP production, while the extracellular acidification rate (ECAR) of glycolysis was slightly increased and glycolysis capacity remained unchanged (Figure S1b, c). These results demonstrate that WD is a disease characterized by abnormal hepatocyte metabolism.

Interestingly, chronic cholecystitis emerged as the most common complication, with a notably higher incidence in WD patients (49.51%) compared to HBV patients (9.46%) in Cohort 2 (**Figure 1e**). Despite the comparable incidence of gallbladder in the two groups and the significantly reduced plasma concentration of total bilirubin in WD patients (Cohort 1 and Cohort 2), this finding holds (**Figure 1e**). Moreover, in cholecystitis patients, the proportion of cases with abnormal index was significantly higher when combined with WD, which included white blood

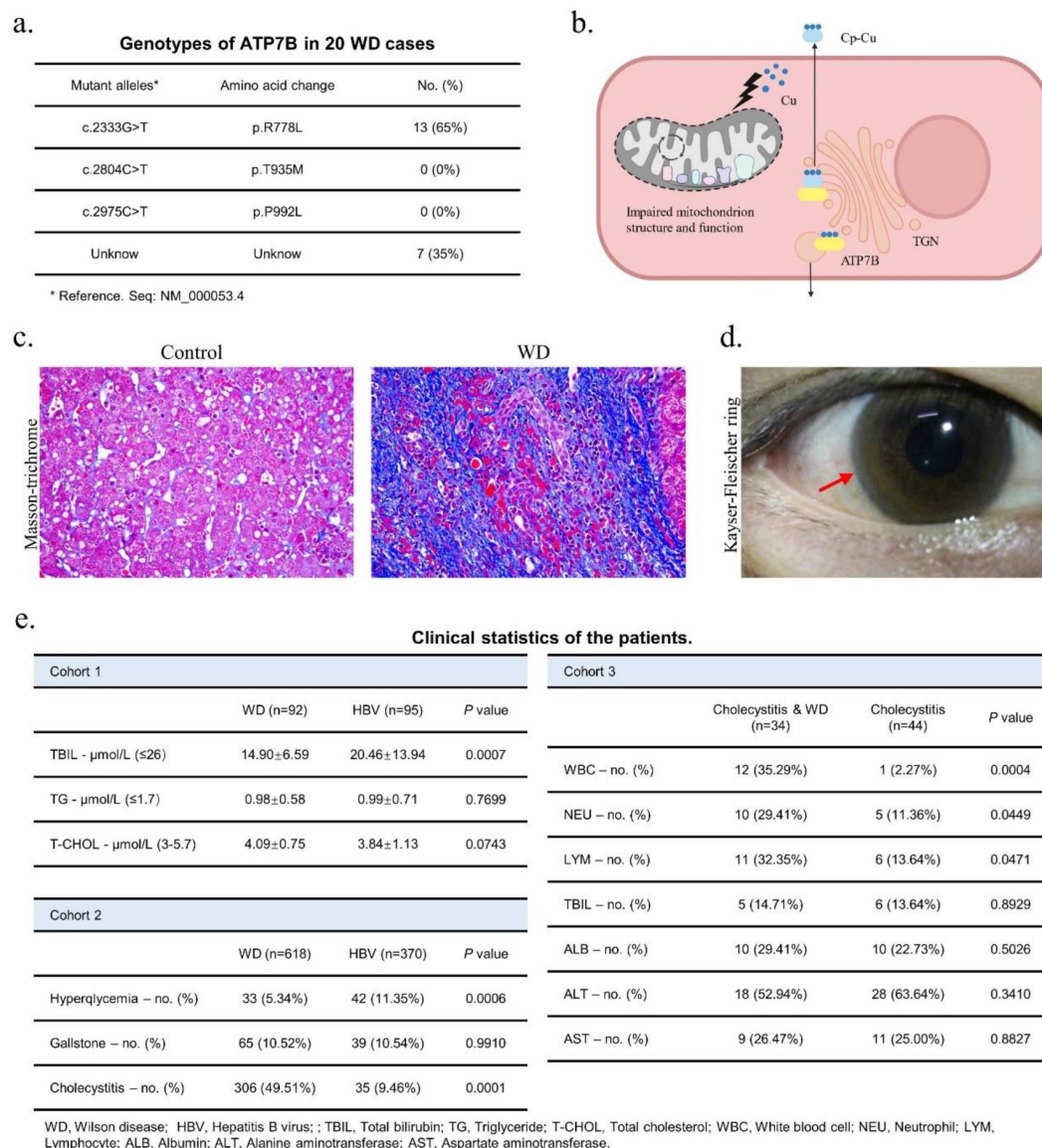


Figure 1.

Clinical manifestation of WD patients and WD causes poor prognosis of cholecystitis. a.

DNA sequencing result of 20 WD patients was summarized in the graph. **b.** The schematic diagram illustrates copper transport disturbances caused by ATP7B mutations, along with structural and functional impairments in mitochondria. Created with BioRender.com. **c.** Masson-trichrome staining of liver tissues of patients with WD and control. **d.** The picture of Kayser-Fleischer (KF) ring in the cornea of a patient with WD, captured during clinical examination. **e.** The table of clinical statistics of patients in 3 cohorts. Data are mean $\pm$ S.D. Unpaired two-tailed t-test and chi-square test.

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cell (WBC), neutrophil (NEU), and lymphocyte (LYM) counts (Cohort 3, [Figure 1e](#)). Overall, these findings suggest that cholecystitis outcomes are negatively affected by WD and that there are immune abnormalities that affect the prognosis of cholecystitis in WD patients.

Single-cell profiling of non-parenchymal cells in the liver of WD patients

The hepatic mesenchymal cells from three WD patients and three liver hemangioma patients were analyzed using the 10X Genomics platform for scRNA-seq ([Figure 2a](#)) to better understand the complicated immune abnormalities in WD patients. 57,384 high-quality single transcriptomes were generated, with 30,068 cells in the case group and 27,316 in the control group. We validated twenty-six principal components via PCA and substantiated the grouping through UMAP dimensionality reduction cluster graph ([Figure 2b, c](#), [Figure S2a](#)). We then utilized individual cell expression profiles and reference to cell marker gene expression to implement an automated annotation method, which was further complemented by manual review and fine-tuning. This initial annotation process resulted in the classification and display of 12 significant cell clusters. The UMAP algorithm was used to segregate and display the clusters. The study identified various cell types, including hepatocytes^{45,46}, cholangiocytes⁴⁵, endothelial cells (ECs)^{47–49}, hepatic stellate cells (HepSCs)^{46,50}, proliferating cells^{48,51,52}, B cells^{53,54}, plasma cells^{52,53}, T and NK cells^{53,55}, neutrophils⁵⁶, mast cells^{48,55,56}, mononuclear phagocytes (MPs)⁵⁷, and plasmacytoid dendritic cells (pDCs)^{48,58} ([Figure 2d, e](#) and [Figure S2b, c](#)). [Figure S2d](#) exhibited the top 10 differently expressed genes (DEGs) for each cell type.

Based on the initial annotation, there were notable variations in the quantities and ratios of different cell types between the case group and control group. Our study observed that T and NK lineages were the primary immune cell population among all the examined samples, accounting for 56.54% in CASE and 73.55% in CON, followed by MPs which constituted 17.33% in CASE and 9.97% in CON ([Figure 2f](#), [Figure S2e](#), and [Supplementary Table S1](#)). The significant increase in the proportion of MPs and decrease in T and NK cells suggest that MPs likely dominate the immune microenvironment remodeling in patients with WD, while lymphocytes play a minor role. B cell proportions (4.53% in CASE and 2.27% in CON), plasma cell proportions (2.36% in CASE and 0.42% in CON), and pDC proportions (0.42% in CASE and 0.16% in CON) were elevated, while neutrophil proportions (3.53% in CASE and 6.45% in CON) declined. The proportion of remaining cells is shown in [Supplementary Table S1](#).

The immune microenvironment was altered in WD patients

To demonstrate dissimilarities in the composition of the immune microenvironment between the two groups of individuals in a more detailed manner, a secondary subtype clustering and annotation process was applied to MPs and T and NK cells ([Figure 3a, b](#), and [Figure S3a, b](#)). MPs are myeloid immune cells that serve phagocytic and antigen-presenting functions. These cells, which include monocytes, macrophages, and dendritic cells, play an essential role in determining the nature of the immune response⁵⁹. A total of 6,962 MPs were identified (4,439 in CASE and 2,523 in CON), and they were subdivided into 7 clusters through annotation ([Figure 3a](#), [Figure S3a](#), and [Supplementary Table S1](#)). Cell marker genes for annotating cell subtypes were depicted in [Figure S3c and d](#), and [Figure S3e](#) demonstrated the top 10 DEGs between the two groups^{46,48,49,51,52,55–58,60–65}. The DEGs were related to various cell types, including proliferating cells, macrophages, monocytes, mature dendritic cells (MatureDCs), conventional type 1 dendritic cells (cDC1), conventional type 2 dendritic cells (cDC2), and Kupffer cells (KCs). In the case group, there was a significant increase in both macrophages (25.03% in CASE versus 14.33% in CON) and KCs, the resident macrophages in the liver (26.03% in CASE versus 22.96% in CON), suggesting their involvement in promoting hepatic inflammation in metabolic liver disease ([Figure 3a](#))⁶⁶. By contrast, the percentages of monocytes (35.45% in CASE and 41.78% in CON), cDC1 (2.14% in CASE and 5.99% in CON), and cDC2 (10.39% in CASE and 14.13% in CON) were decreased in the case group ([Figure 3a](#)).

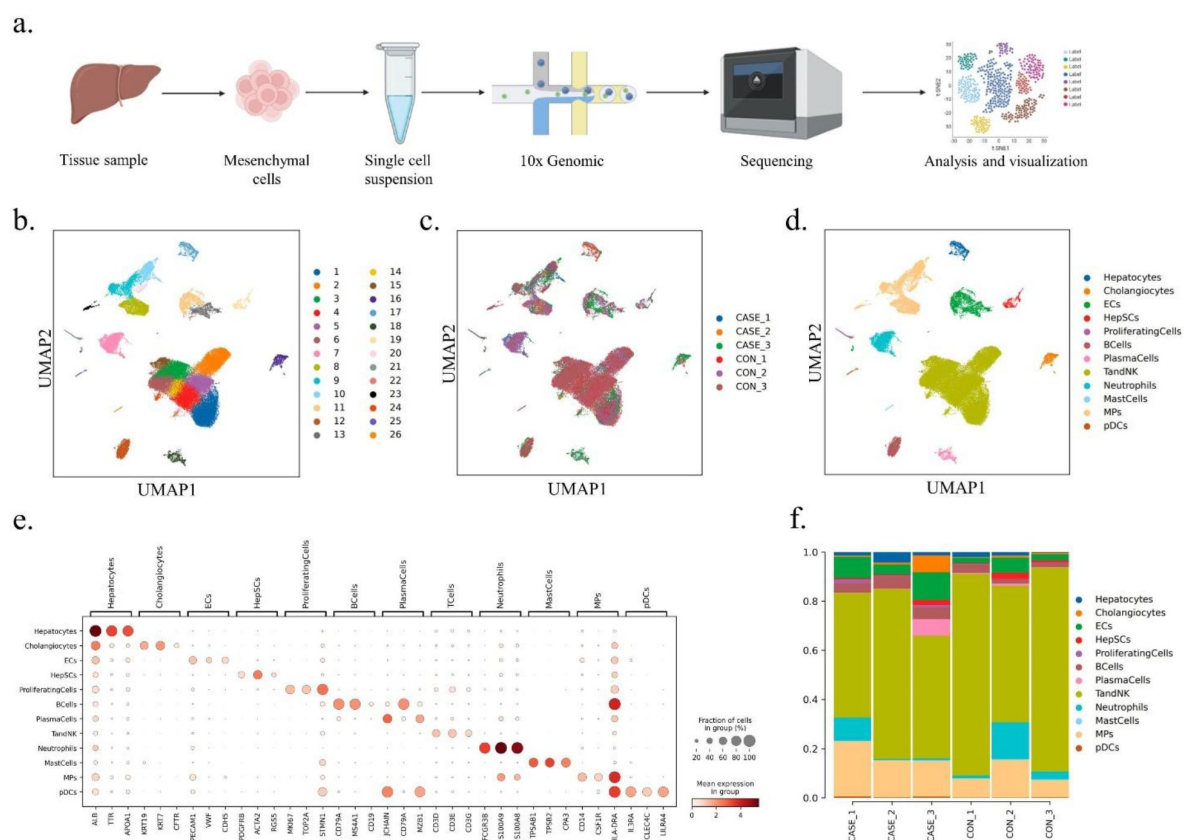


Figure 2.

Single-cell profiling of non-parenchymal cells in the liver of WD patients. a.

The workflow of scRNA-seq. Created with BioRender.com. **b.** UMAP visualization of 26 principal components of all the single cells. **c.** UMAP visualization of 26 principal components by samples. **d.** UMAP visualization of 12 main cell types. **e.** The dot plot showing the expression of top 3 marker genes in 12 main cell types.

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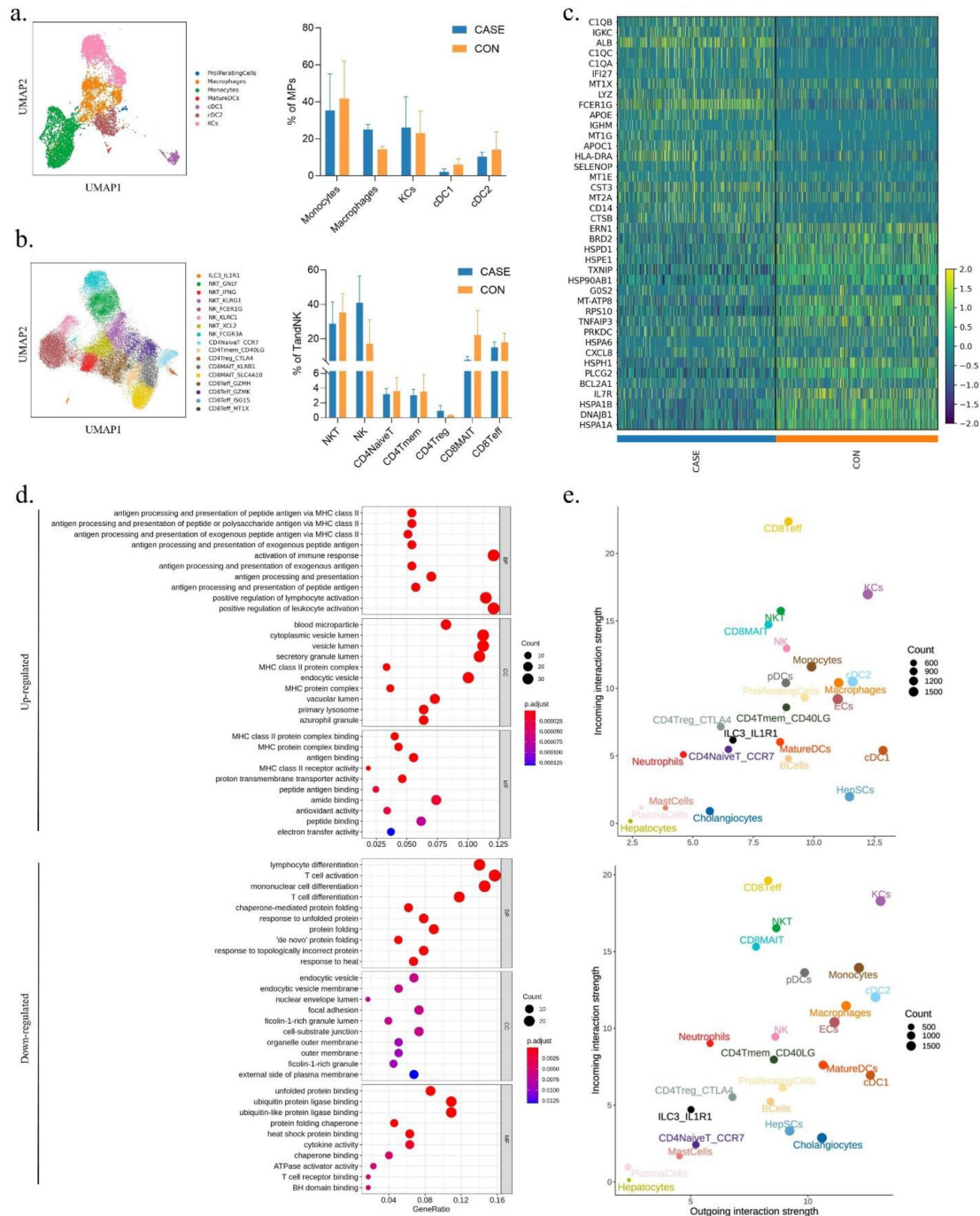


Figure 3.

The altered immune microenvironment in WD patients. a.

UMAP visualization of MPs colored and labeled by cell subtypes (left). The bar chart showing the average percentage of different subtypes in case group (n=3) and control group (n=3) (right). Data are mean $\pm$ S.D. **b.** UMAP visualization of T and NK cells colored and labeled by cell subtypes (left). The bar chart showing the average percentage of different subtypes in case group (n=3) and control group (n=3) (right). Data are mean $\pm$ S.D. **c.** The heatmap showing the top 20 differently expressed genes (DEGs) for all immune cells. **d.** The dot plot showing GO enrichment on DEGs of all immune cells between the case and control group. BP, biological processes; CC, cellular components; MF, molecular function. **e.** The scatter showing the strength of all cell subtypes in signal incoming and outgoing. Upper, the case group; down, the control group.

Seventeen clusters were obtained through reclustering for the largest population of T and NK cells, totaling 34,339 cells (15,403 cells in CASE and 18,936 in CON)^{53,56,67,82}. The clusters included ILC3_IL1R1, NKT_GNLY, NKT_IFNG, NKT_KLRG1, NK_FCER1G, NK_KLRG1, NKT_XCL2, NK_FCGR3A, CD4NaiveT_CCR7, CD4Tmem_CD40LG, CD4Treg_CTLA4, CD8MAIT_KLRB1, CD8MAIT_SLC4A10, CD8Teff_GZMH, CD8Teff_GZMK, CD8Teff_ISG15, and CD8Teff_MT1X (**Figure 3b**, Figure S3b, Supplementary Table S1). The proportion of NK cells (40.97% and 17.07% in CASE and CON, respectively) and CD4Treg (0.92% and 0.31% in CASE and CON, respectively) was significantly increased in the case group. On the other hand, a decrease of NKT cells (28.72% in CASE and 35.26% in CON), CD8MAIT (7.76% in CASE and 22.14% in CON) and CD8Teff (14.95% in CASE and 17.88% in CON) was observed in the case group. CD4 regulatory T cells are thought to inhibit T cell activation in autoimmune and virus-related liver diseases, indicating a correlation between elevated CD4 regulatory T cells and reduced CD8 T cells in WD patients⁸³. The distribution of other types of cells between the two groups is shown in **Figure 3b** and Supplementary Table S1.

To systematically evaluate the key functional alteration, we analyzed the top 20 differently expressed genes in all immune cell populations. We found that the expression of genes encoding complement C1q (*C1QA*, *C1QB*, and *C1QC*) and *HLA-DRA* was significantly upregulated in the case group. Conversely, the expression of genes encoding heat shock protein (*HSPA1A*, *HSPA1B*, *HSPM1*, *HSPA6*, *HSP90AB1*, *HSPE1*, and *HSPD1*) was markedly downregulated in the case group (**Figure 3c**). The Gene Ontology (GO) analysis results demonstrate that WD enhances the abilities of antigen-presenting cells to activate immune response and related lymphocytes, through upregulating antigen processing and presentation, activation of immune response, and positive regulation of lymphocyte activation. However, heat shock proteins delivering antigens may be responsible for the downregulation of lymphocytes differentiation and T cell activation⁸⁴. Next, we analyzed the cellular communication network using Cellchat. Our findings reveal that CD8Teff, KCs, NKT, CD8MAIT, and monocytes were the primary partners in intercellular communication (**Figure 3e**). Furthermore, CD8Teff and NK cells showed stronger signals in the case group, while neutrophils exhibited weaker signals, in both outgoing and incoming (**Figure 3e**). Collectively, these results demonstrate that the immune microenvironment in the liver of WD patients is remodeled, with an obvious enhancement in both quantity and function of innate immunocytes.

The dysfunction of main immune cells in WD patients

To obtain a detail analysis of function comparison in immune cells between two groups, we scored and evaluated the core functional signature of major cell types. We further clustered macrophages, which accounted for 14.33% of the MPs in the control group and increased to 25.03% in the case group, resulting in the identification of four distinct subtypes (Figure S4a). Interestingly, these four subtypes in the patients with WD exhibited distinct gene expression profiles. Macrophages_3 in WD displayed elevated expression levels of S100A family genes (*S100A8*, *S100A9*, *S100A12*, *S100A6*, *VCAN*, *FCN1*, and *S100A4*), indicating a potential contribution to MDSC-like macrophages. In contrast, Macrophages_4 in WD were characterized by upregulation of lipid metabolism genes (*FABP5*, *APOC1*, *CTSD*, and *PLA2G7*) (Figure S4b)^{85,86}. By scoring various gene signatures, we compared the M1 polarization, M2 polarization, pro-inflammatory signature, and interferon response signature of four subtypes within two groups (**Figure 4a**). These four subtypes displayed consistently overlapped M1/M2 signatures. Significantly, the Macrophages_3 subtype exhibited a more pronounced pro-inflammatory signal in the case group, while the other subtypes showed a slight decrease. In addition, the interferon response signature showed slight elevation in four subtypes within the case group. Kupffer cells, which are liver-resident macrophages with unique functions in maintaining homeostasis, represent the most abundant population of tissue-resident macrophages in the human body⁶⁶. KCs were also further clustered and three subtypes were obtained (Figure S4c). Significant differential expression of genes related to the inflammatory response, including *IL1B*, *CXCL2*, *CCL3C1*, *CXCL8*, and *CCL2*, were identified between the two groups (Figure S4d). Similarly, KCs subtypes in two

groups also showed overlapping M1/M2 signatures, but all subtypes in the case group showed varying degrees of increase in pro-inflammatory and interferon-responsive signatures (**Figure 4b** [↗](#)).

Neutrophils, the most abundant leukocytes in human blood, respond to inflammatory stimuli by migrating to the site of inflammation. Its immunological functions, such as phagocytosis, reactive oxygen species generation, degranulation, and production of cytokines and chemokines, play a crucial role in the resolution of inflammation and tissue repair^{87,88} [↗](#). In the case group, a lower percentage of neutrophils (3.53% in CASE and 6.45% in CON) was observed, and the functional characteristics were subsequently evaluated. The results showed that the maturation, chemotaxis, phagocytosis, and chemokine activity scores were decreased in the CASE group, while the activation of the type I interferon signaling pathway was enhanced (**Figure 4c** [↗](#)).

Since the biological process of antigen processing and presentation was upregulated, we calculated the relative expression intensity of genes encoding MHC molecules and complement molecules in cDC1, cDC2, and mature DCs (**Figure 4d** [↗](#)). Notably, genes encoding MHC class I molecules (*HLA-A*, *-B*, *-C*, *-E*, and *-F*) were significantly upregulated in the mature DCs in the case group, while genes encoding MHC class II molecules (*HLA-DQA1*, *-DQA2*, *-DQB1*, *-DQB2*, *-DPA1*, *-DPB1*, *-DRA*, *-DRB1*, and *-DRB5*) were all downregulated in the mature DCs. Monocytes accounted for the highest proportion (35.45% in CASE and 41.78% in CON), and most of them were CD14⁺ classical monocytes (Figure S4e). Compared to the control group, *S100A8*, *S100A9*, and *S100A12* were the top 3 upregulated genes in classical monocytes of WD patients (Figure S4f). *S100A8* and *S100A9* are confirmed to play vital roles in modulating the inflammatory responses by recruiting leukocytes and inducing cytokine secretion during inflammation, and as an alarmin, their elevation indicates inflammation in the local environment^{89,90} [↗](#). GO analysis of DEGs in monocytes showed a decrease in biological processes involving cytokine production, response to lipopolysaccharide and bacteria, as well as lymphocyte activation (**Figure 4e** [↗](#)).

For T cells, we performed GO analysis on DEGs in CD8 T cells between two groups. Notably, the biological processes linked to energy and lipid metabolism were upregulated in several subtypes, including CD4Tmem_CD40LG, CD4Treg_CTLA4, CD8MAIT_KLRB1, CD8MAIT_SLC4A10, CD8Teff_GZMH, and CD8Teff_GZMK. In addition, biological processes associated with response to copper ion were upregulated in the CD8Teff_MT1X subtype (Figure S4g). Meanwhile, T cell subtypes showed comparable cytotoxicity scores, and there were slight changes between two groups (**Figure 4f** [↗](#)). However, CD8 T cells in the case group exhibited lower levels of exhaustion, except for CD8Teff_GZMH and CD8Teff_ISG15 (**Figure 4f** [↗](#)). It was also discovered that while the proportion of CD4Treg_CTLA4 was increased in the case group, the Treg signature was slightly lower (**Figure 4g** [↗](#)). Therefore, these results suggest that WD patients exhibit variations in immune cell function and certain biological processes.

NK cell exhaustion in WD patients

The KEGG analysis of DEGs in NK cells between the case and control groups revealed down-regulation of the natural killer cell-mediated cytotoxicity, along with the NF-kappa B signaling pathway and chemokine signaling pathway, in the case group (Figure S5a). Interestingly, GO analysis showed upregulated biological processes of oxidative phosphorylation and ATP metabolic process, implying the altered metabolic program in NK cells and immune microenvironment (Figure S5b). Since the primary effect of NK cells is cytotoxicity on target cells, we compared the gene expression levels of activating receptors, inhibitory receptors, and effector molecules between two groups. The results showed increased expression in *KLRC1* and *TIGIT*, and decreased expression in *FCGR3A*, *TNF*, *IFNG*, *GZMB*, *GNLY*, and *CCL4* (**Figure 5a** [↗](#)). *KLRC1* and *TIGIT* are inhibitory receptors expressed on NK cell surface, and their activation inhibits NK cell functions²¹ [↗](#),⁹¹ [↗](#). It was collectively indicated that WD patients have NK cells with exhausted status, which results in poor effector function.

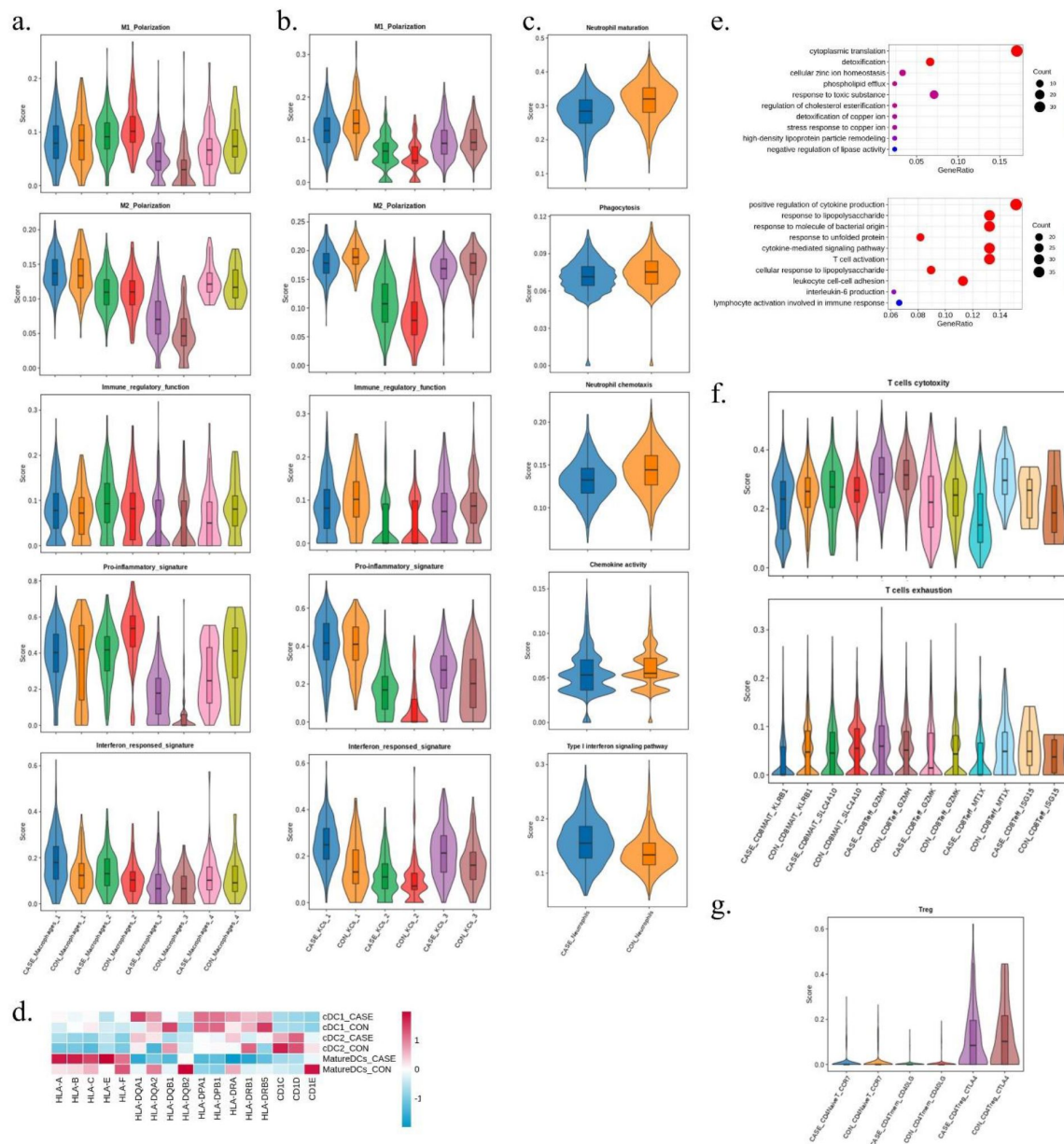


Figure 4.

The dysfunction of main immune cells in WD patients. a.

The violin plots displaying M1 polarization, M2 polarization, pro-inflammatory signature, and interferon responded signature of the 4 subtypes within macrophages between two groups by gene scoring. **b.** The violin plots displaying M1 polarization, M2 polarization, pro-inflammatory signature, and interferon responded signature of the 3 subtypes within Kupffer cells between two groups by gene scoring. **c.** The violin plots displaying maturation, phagocytosis, chemotaxis, chemokine activity and Type I interferon signaling pathway scores within neutrophils between two groups. **d.** The heatmap showing the relative expression level of genes in each cell subtype between the case and control group. **e.** The dot plot showing biological process of GO enrichment on differentially expressed genes (DEGs) in monocytes between the case and control group. Upper, upregulation; down, downregulation. **f.** The violin plots displaying cytotoxicity and exhaustion scores of the 6 cell subtypes within CD8 T cells between two groups by gene scoring. **g.** The violin plots displaying Treg scores of the 3 cell subtypes within CD4 T cells between two groups by gene scoring.

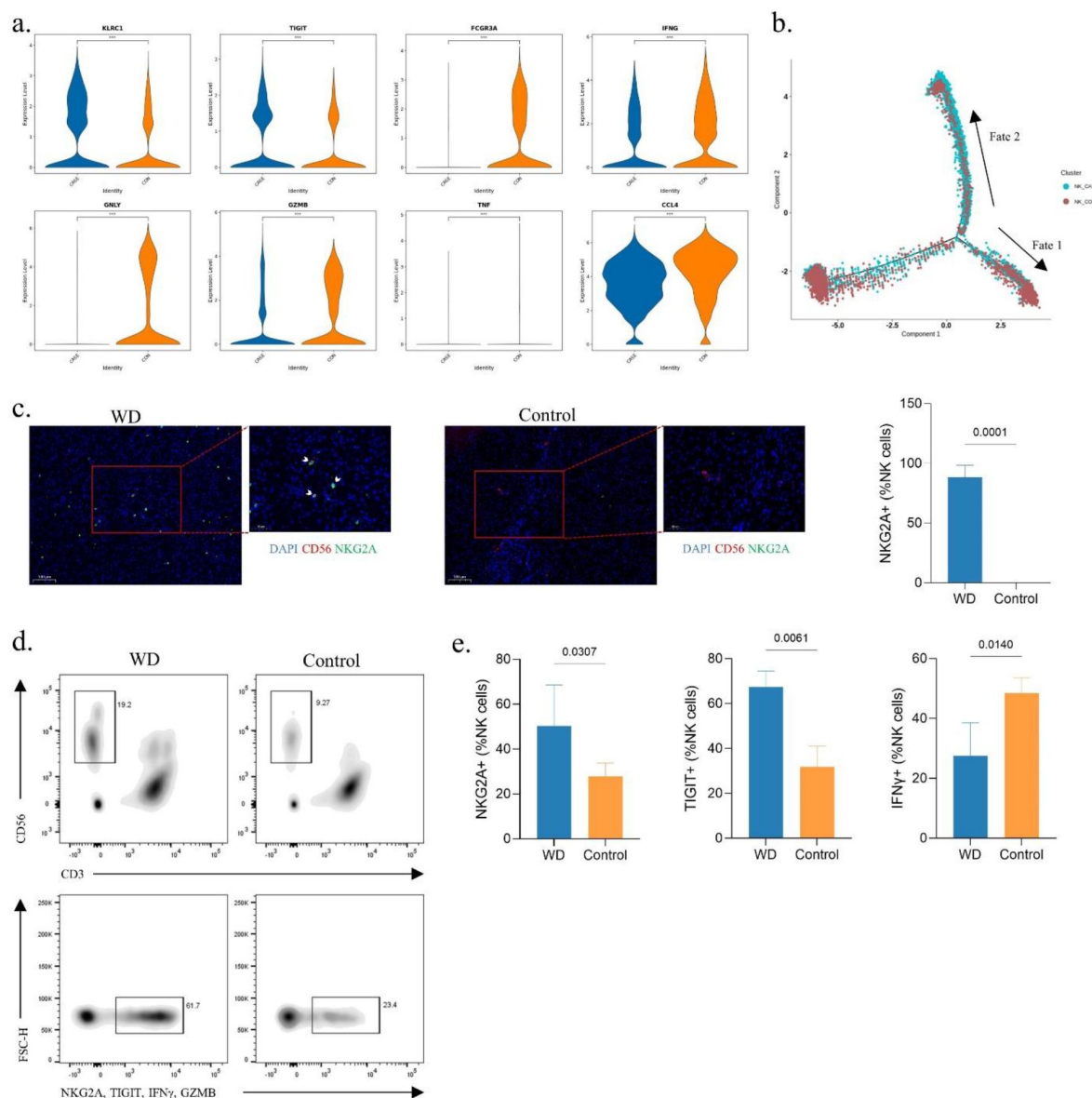


Figure 5.

Identification of NK cell exhaustion in WD patients. a.

The violin plots showing the relative expression level of genes in NK cells between the case and control group. *******, $P < 0.001$; Wilcoxon test. **b.** Trajectories of NK cells in the case and control group. **c.** Representative immunofluorescence staining for NK cells (CD56) and expression of NKG2A in liver tissues of WD and control. Scale bars, 100 μ m for 40X and 50 μ m for 100X. NKG2A⁺ NK cells are highlighted by arrows. Quantifications were performed by assessing three random fields per slide. Data are mean $\pm$ S.D. Unpaired two-tailed t-test. **d.** Gating strategy applied in flow cytometry. **e.** The bar charts showing the percentage of NKG2A⁺ (WD, $n = 5$; Control, $n = 5$), TIGIT⁺ (WD, $n = 3$; Control, $n = 3$), and IFN γ ⁺ (WD, $n = 4$; Control, $n = 4$) NK cells. Data are mean $\pm$ S.D. Unpaired two-tailed t-test.

In addition, trajectory analysis showed that NK cells in two groups exhibited a distinct differentiation trajectory. Most of the NK cells in the control group differentiated towards cell fate 1, while those in the case group displayed a higher tendency for differentiation towards cell fate 2, with cell grouping occurring at the final stage (**Figure 5b**). To ascertain the gene expression profile and the respective cell statuses of the two outcomes, we conducted pseudotime analysis of gene expression in two differentiating directions. The results showed that the gene expression of *KLRC1* (included in gene cluster 1) was gradually increased along the way to cell fate 2, while the gene expression of *FCGR3A*, *TNF*, and *GZMB* (included in gene cluster 3) was weakened in the same direction, suggesting that cell fate 2 is a more exhausted state of NK cells (Figure S5c). Consequently, these results demonstrate that NK cells differentiate towards an exhausted status in WD patients. The development, maturation, and function of NK cells are strictly regulated by transcription factors (TFs)⁹². Next, to assess the transcription factors (TFs) underlying differences in NK cells between WD patients and control groups, we applied PYSCEPIC analysis in NK cells. We found that the gene expression of T-bet (encoded by *TBX21*) was downregulated and Eomes (encoded by *EOMES*) was upregulated in the case group (Figure S5d). These two constitute a pair of TFs that play significantly complementary roles in directing the transcriptional program of NK cells. Consistent with previous studies, T-bet expression was downregulated in exhausted NK cells, whereas Eomes expression was increased^{24,93,94}. Thus, it was suggested that the downregulation of *TBX21* and upregulation of *EOMES* contributed to the exhaustion of NK cells in WD patients.

To validate NK cell exhaustion in WD patients, we performed multiplex immunohistochemistry (mIHC) staining on liver tissue sections. The results indicate that NKG2A⁺ cells were more frequently observed in CD56⁺ NK cells in WD patients (**Figure 5c**). Moreover, flow cytometry also confirmed a significant increase in the percentage of NKG2A⁺ and TIGIT⁺ NK cells and a significant decrease in the percentage of IFN γ ⁺ NK cells in peripheral blood lymphocytes from WD patients compared to control groups (**Figure 5d, e**). Meanwhile, the proportion of NK cells in lymphocytes from peripheral blood and the proportion of CD56^{Bright} NK cells in total NK cells were also calculated. The significantly higher proportion of NK cells in WD patients was consistent with the scRNA sequencing data, and more CD56^{Bright} NK cells suggested a relatively weaker effector function in WD patients (Figure S5e). Therefore, these results indicated that, despite the increased proportion and the upregulated oxidative phosphorylation, NK cell exhibited exhausted status in WD patients.

NK cell exhaustion predicts poor prognosis of cholecystitis

Cholecystitis patients with WD exhibit poorer clinical outcomes. There is evidence demonstrating that NK cell exhaustion is correlated with poor liver cancer prognosis^{24,26,95}. Thus, we hypothesize that NK cell exhaustion also contributes to the worse prognosis of cholecystitis patients with WD.

Prognostic models for inflammatory related diseases in GEO were developed using machine learning methods, including random survival forest, ROC curve, and Kaplan-Meier survival curves. A logistic regression algorithm was used to construct a multigene prediction model based on GSE166915 (cholecystitis) and other inflammatory diseases, namely GSE91035 (pancreatitis), GSE75037 (lung adenocarcinoma), and GSE41804 (hepatocellular carcinoma). Using stepwise regression analysis, 12 genes, *KLRC1*, *TIGIT*, *HAVCR2*, *PDCD1*, *CD96*, *LAG3*, *FCGR3A*, *NCR3*, *NCR2*, *NCR1*, *KLRC1*, and *PRF1*, were selected to obtain the best model. The results showed that the predictive model constructed from these genes had good prognostic performance, with AUC of 0.77, 0.80, 0.97, 0.90 (Figure S6a-d). The AUCs of the models in GSE75037 and GSE41084 were relatively high, 0.97 and 0.90, respectively. Univariate Cox regression analysis revealed that the *TIGIT* (in GSE166915) and *PDCD1* (in GSE75037) were risk genes with hazard ratio (HR) > 1, while *NCR2* (in GSE91035), *HAVCR2*, *KLRC1*, *NCR3* (in GSE75037), *NCR3*, *LAG3* (in GSE41804) were protective gene with HR < 1 among 12 differently expressed markers of NK cell exhaustion (Figure S6a-d). Consistently, the Kaplan-Meier survival curves suggested a shorter survival time in the

group with high markers of NK cell exhaustion than in the group with low markers in the GSE166915 ($P= 6.1\text{e-}6$), GSE91035 ($P= 1.9\text{e-}5$), GSE75037 ($P= 3.8\text{e-}16$), and GSE41804 ($P= 3.0\text{e-}3$) datasets (**Figure 6a-d**). Patients with cholecystitis, pancreatitis, lung adenocarcinoma, and hepatocellular carcinoma who show high markers of NK cell exhaustion are associated with poor survival compared to those with low markers of NK cell exhaustion.

Discussion

Immune cell dysfunction impacts the development, progression, and prognosis of diseases and serves as the basis for immunotherapy. The current understanding of immune cell dysfunction in inflammatory diseases is limited, which hinders the development of effective therapeutic strategies. Our retrospective clinical analysis indicates that WD contributes to the unfavorable prognosis of cholecystitis. We developed an atlas of immune cell dysfunction linked to specific metabolic disruption in WD patients, including 19 major cell types. The immune microenvironment change primarily encompasses increased proportion and strengthened function in innate immunocytes, including upregulated antigen processing and presentation and activation of immune responses. Furthermore, we conducted a comprehensive assessment and comparison of immune cell function in WD and control group. This includes an evaluation of pro-inflammatory and immune regulatory function in macrophages and KCs, phagocytosis and chemokine activity in neutrophils, and cytotoxicity in T cells, etc. We identified NK cell exhaustion in the WD group, with upregulation of *NKG2A* and *TIGIT* and downregulation of *CD16*, *TNF*, *IFNG*, *GZMB*, *GNLY*, and *CCL4*. The validation of our findings in patient liver tissues and blood samples was conducted using immunohistochemistry assays and flow cytometry. Our findings indicated that the NK cell exhaustion signature has the ability to predict and prognosticate inflammatory diseases such as cholecystitis, pancreatitis, lung adenocarcinoma, and hepatocellular carcinoma. These results unveil novel perspectives on the immune cell dysfunction role in inflammatory diseases.

WD represents a model of metabolic abnormality in hepatocytes caused by *ATP7B* mutation. Therefore, immune cell dysfunction was induced by the extracellular signals within the reshaped tissue microenvironment. Emerging research indicates that metabolic crosstalk in the tumor microenvironment disrupts metabolic and signaling pathways in immune cells, leading to dysfunction^{96–100}. Targeting metabolic pathways to enhance the efficacy of immunotherapy is a promising approach. Our study found multiple metabolic abnormalities in immune cells, including the widely upregulated ATP metabolic process in T and NK cells. Interestingly, glycolysis and oxidative phosphorylation contribute to maintain the cytotoxic ability and promote expansion of NK cells^{101–103}. However, high rate of oxidative phosphorylation, increased proportion, and exhausted status were simultaneously observed in NK cells of WD patients. Further studies are needed to elucidate the metabolic landscape, mechanisms underlying metabolic alterations, and their role in immune cell dysfunction. Defective *ATP7B* leads to excessive copper accumulation in the system's circulation. A recent study showed that excess copper induces cell death in a manner dependent on mitochondrial respiration¹⁰⁴. Considering the impact on mitochondrial respiration processes in immune cells, it is worth investigating the role of copper in the physiology of immune cells. Further research is expected to establish a foundation for improving immunotherapies through metabolic or ionophore interventions.

Methods

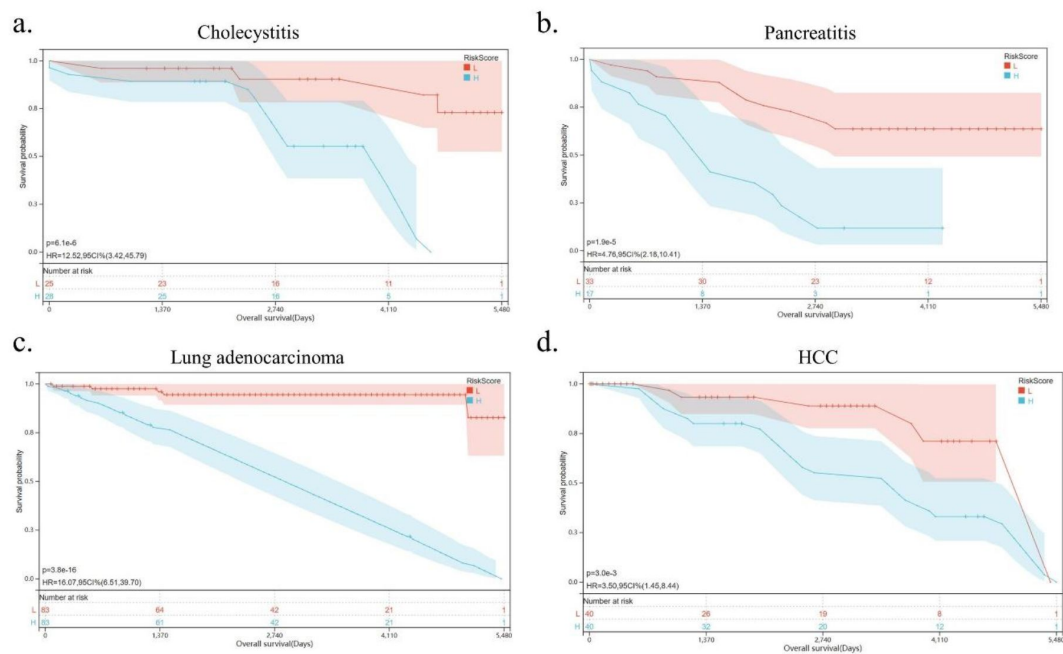


Figure 6.

NK cell exhaustion predicts poor prognosis of inflammatory diseases. a.-d.

Kaplan-Meier survival curves for cholecystitis, pancreatitis, lung adenocarcinoma, and hepatocellular carcinoma according to the markers of NK cell exhaustion in the GSE166915, GSE91035, GSE75037, and GSE41804 datasets.

Patients and clinical samples

All the clinical data were obtained from The First Affiliated Hospital of Anhui University of Chinese Medicine. The whole blood samples and liver tissue samples were collected from patients at The First Affiliated Hospital of Anhui University of Chinese Medicine. The study was approved by the Ethics Committee of the First Affiliated Hospital of Anhui University of Traditional Chinese Medicine and the Ethics Committee of the First Affiliated Hospital of USTC.

Sanger sequencing

The sanger sequencing on 20 WD patients were conducted by Sangon Biotech (Shanghai) Co., Ltd. The primers to detect mutations are:

R778L-F, 5'-AGCCTTCACTGTCCTTGTCTTTC-3';

R778L-R, 5'-ATTAGCTGGGATTTCAGAAGTAGTG-3';

T935M-F, 5'-ATGCTTGTGGTGTTCCTTCTTC-3';

T935M-R, 5'-AGAAGCAAGCAAATAAAATGTAATG-3';

P992L-F, 5'-TCTCAACCTGCCTCTGACTCTG-3';

P992L-R, 5'-TGGTGGCTACTCTGTTGCTACTG-3'.

Cell culture, gene knockout, and metabolic assays

WRL 68 cells were cultured using DMEM high-glucose media supplemented with 10% FBS at 37 °C with 5% CO₂. The lentivirus-mediated CRISPR-Cas9 KO vector targeting human *ATP7B* was obtained from OBiO Technology (Shanghai) Co., Ltd. Cells at 30% to 40% confluency were incubated in a medium containing optimal dilutions of lentivirus, then subjected to puromycin selection to obtain the stable transfected cells. OCRs and ECARs were measured using Seahorse XF Cell Mito Stress Test Kit (103015-100, Agilent) and Seahorse XF Glycolysis Stress Test Kit (103020-100, Agilent) according to manuals in Agilent XFe96 Analyzer, respectively.

Single-cell RNA sequencing

In the cell preparation section: For the quality check and counting of single cell suspension, the cell survival rate is generally above 80%. The cells that have passed the test are washed and resuspended to prepare a suitable cell concentration of 700~1,200 cells/μL for 10x Genomics Chromium™. The system is operated on the machine.

Steps in GEM Creation & Thermal cycling: GEMs (Gel Bead in Emulsion) were constructed for single cell separation according to the number of cells to be harvested. After GEMs were normally formed, GEMs were collected for reverse transcription in a PCR machine for labeling.

Post Cycling Cleanup & cDNA Amplification: The GEMs were oil-treated, and the amplified cDNA was purified by magnetic beads, and then subjected to cDNA amplification and quality inspection.

Library Preparation & Quantification: The 3' Gene Expression Library was constructed with the quality-qualified cDNA. After fragmentation, adaptor ligation, sample index PCR, etc., the library is finally quantitatively examined.

Sequencing: The final library pool was sequenced on the Illumina HiSeq instrument using 150-base-pair paired-end reads.

Cell type annotation and differentially expressed genes (DEGs) analysis

The cell type identity of each cluster was determined with the expression of canonical markers found in the DEGs using SynEcoSys database. The markers and references are shown in Supplementary Table S4. To identify DEGs, we used the Seurat FindMarkers function based on Wilcox likelihood-ratio test with default parameters, and selected the genes expressed in more than 10% of the cells in a cluster and with an average log (Fold Change) value greater than 0.25 as DEGs. For the cell type annotation of each cluster, we combined the expression of canonical markers found in the DEGs with knowledge from literatures, and displayed the expression of markers of each cell type with heatmaps that were generated with Seurat Heatmap function. Doublet cells were identified as expressing markers for different cell types, and removed manually.

Pathway enrichment analysis

To investigate the potential functions of DEGs, the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis were used with the “clusterProfiler” R package 3.16.1¹⁰⁵. Pathways with p_adj value less than 0.05 were considered as significantly enriched. Gene Ontology gene sets including molecular function (MF), biological process (BP), and cellular component (CC) categories were used as reference.

Cell-cell Interaction Analysis

CellChat (version 0.0.2) was used to analyze the intercellular communication networks from scRNA-seq data¹⁰⁶. A CellChat object was created using the R package process. Cell information was added into the meta slot of the object. The ligand-receptor interaction database was set, and the matching receptor inference calculation was performed.

Trajectory analysis

Cell differentiation trajectory was reconstructed with Monocle2¹⁰⁷. Highly-variable genes (HVGs) were used to sort cells in order of spatial-temporal differentiation. We used DDRTree to perform FindVariableFeatures and dimension-reduction. Finally, the trajectory was visualized by plot_cell_trajectory function. Next, CytoTRACE (a computational method that predicts the differentiation state of cells from single-cell RNA-sequencing data using gene Counts and Expression) was used to predict the differentiation potential¹⁰⁸.

Transcription factor regulatory network analysis

Transcription factor network was constructed by pycenic v0.11.0 using scRNA expression matrix and transcription factors in AnimalTFDB¹⁰⁹. First, GRNBoost2 predicted a regulatory network based on the co-expression of regulators and targets. CisTarget was then applied to exclude indirect targets and to search transcription factor binding motifs. After that, AUCell was used for regulon activity quantification for every cell.

UCell gene set scoring

Gene set scoring was performed using the R package UCell v 1.1.0¹¹⁰. UCell scores are based on the Mann-Whitney U statistic by ranking query genes in order of their expression levels in individual cells. Because UCell is a rank-based scoring method, it is suitable to be used in large datasets containing multiple samples and batches. The gene sets for signature scoring are listed in Supplementary Table S3.

Multiplex immunohistochemistry

The procedure of deparaffinization, dehydration, antigen retrieval, endogenous peroxidases quench, block, antibody incubation was conducted as standard immunohistochemistry. Solution of tyramide signal amplification (TSA) (WAS1101010, BioMed World) was applied on the slides and antibodies were removed by microwave heating. Repeat block, antibody incubation, TSA, and microwave heating to perform next cycle of staining. DAPI (WAS1301050, BioMed World) was used to stain nuclei and antifade mountant (WAS1302050, BioMed World) was applied. The primary antibody to CD56 (ab75813, Abcam) and NKG2A (ab260035, Abcam) and second antibody (WAS1201100, BioMed World) were used. Image of immunofluorescence-stained slides were acquired with Nikon Eclipse Ci-L and 3DHISTECH Panoramic MIDI II.

Flow cytometry

Lymphocytes were isolated from the whole blood using density gradient centrifugal method (P8610, Solarbio). Isolated cells were blocked (564219, BD Pharmingen) and stained with antibodies against CD3 (560835, BD Pharmingen), CD56 (555518, BD Pharmingen), NKG2A (375114, Biolegend), and TIGIT (568672, BD Pharmingen). For intracellular cytokine staining, cells were stimulated for 4 h with Leukocyte Activation Cocktail and BD GolgiPlug (550583, BD Pharmingen) according to manuals. After stimulation, cells were blocked, stained for surface markers, fixed and permeabilized with Fixation/Permeabilization Kit (554714, BD Pharmingen). Fixed cells were stained with antibodies against IFN γ (552887, BD Pharmingen).

Construction and verification of prognostic model

On the basis of the markers of NK cell exhaustion, univariate Cox analysis was conducted in cholecystitis, pancreatitis, lung adenocarcinoma, and hepatocellular carcinoma patients to screen survival-related markers of NK cell exhaustion, and markers with P values < 0.05 were retained. We conducted multivariate analysis to identify the optimal prognostic markers for the prognostic model. The risk scores of the patients were calculated based on the normalized gene expression levels and the Cox regression coefficients of selected markers. Time-dependent receiver operating characteristic (ROC) curves were utilized to verify the prognostic performance of the model for overall survival (OS). For Kaplan-Meier survival, we used the R software package maxstat (Maximally selected rank statistics with multiple p-value approximations version: 0.7-25) to calculate the optimal cutoff value for RiskScore. We set the minimum group sample size to be greater than 25% and the maximum group sample size to be less than 75%, ultimately obtaining the optimal cutoff value. Based on this, patients were divided into high and low groups, and further analyzed the prognostic differences between the two groups using the R software package's survival function. The logrank test method was used to evaluate the significance of prognostic differences between different groups of samples, and significant prognostic differences were ultimately observed.

Statistical analysis

Statistical analyses were performed using appropriate tests as indicated in legends (unpaired two-tailed t-test, chi-square test, and Wilcoxon test).

Data availability

sc-RNA sequencing data that support the findings of this study have been deposited in Gene Expression Omnibus (GEO) with the accession numbers GSE254082.

Author contributions

Y.W., Q.F., H.W., and Q.Y. conceived the concept for this study. Y.J., J.X., L.J., and M.H. analyzed data and implemented experiments. C.D. conducted the bioinformatic analysis. L.J., Q.T., and Z.L. collected the clinical data and clinical samples. Y.J., J.X., and C.D. wrote the manuscript. W.Z. and Q. F. proofread the manuscript.

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Competing interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

Summary:

Wilson's Disease (WD) is an inherited rare pathological condition due to a mutation in ATP7B that alters mitochondrial structure and dysfunction. Additionally, WD results in dysregulated copper metabolism in patients. These metabolic abnormalities affect the functions of the liver and can result in cholecystitis. Understanding the immune component and its contribution to WD and cholecystitis has been challenging. In this work, the authors have performed single-cell RNA sequencing of mesenchymal tissue from three WD patients and three liver hemangioma patients.

Strengths:

The authors describe the transcriptomic alterations in myeloid and lymphoid compartments.

Weaknesses:

In brief, this manuscript lacks a clear focus, and the writing needs vast improvement. Figures lack details (or are misrepresented), the results section only catalogs observations, and the discussion needs to focus on their findings' mechanistic and functional relevance. The major weakness of this manuscript is that the authors do not provide a mechanistic link between the absence of ATP7B and NK cells' impaired/altered functions. While the work is of high clinical relevance, there are various areas that could be improved.

<https://doi.org/10.7554/eLife.98867.1.sa1>

Reviewer #2 (Public Review):

Summary:

Wilson's disease is a rare genetic disorder caused by mutations in the ATP7B gene. Previous studies have documented that ATP7B mutations can disrupt copper metabolism, affecting brain and liver function. In this paper, the authors performed a retrospective clinical study and found that Wilson's disease has a high incidence of cholecystitis. Single-cell RNA-seq analysis revealed changes in the immune microenvironment, including the activation of immune responses and the exhaustion of natural killer cells.

Strengths:

A key finding of this study is that the predominant ATP7B gene mutation in the Chinese population is the 2333G>T (p. R778L) mutation. The authors reported associations between Wilson's disease and cholecystitis, as well as the exhaustion of natural killer cells.

Weaknesses:

The underlying mechanisms linking ATP7B mutations to cholecystitis and natural killer cell exhaustion remain unclear. Specifically, it is not yet determined whether copper metabolism alterations directly cause cholecystitis and natural killer cell exhaustion, or if these effects are secondary to liver dysfunction.

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